

Modeling and Understanding Nurse Stress Using Wearable Physiological Sensor Data

Sumeyye Adanur

Table of Contents

1	<i>Problem Statement</i>	3
2	<i>Methodology</i>	3
2.1	<i>Data Preprocessing</i>	3
2.1.1	Motion Features Engineering	3
2.1.2	EDA Signal Processing	3
2.1.3	Heart Rate (HR) Feature Engineering	4
2.1.4	Temporal Feature Construction	4
2.1.5	Handling Irregular Sampling and Resampling	5
2.2	<i>Modeling</i>	6
3	<i>Results and Evaluation</i>	7
3.1	<i>Results</i>	7
3.2	<i>Evaluation</i>	7
3.3	<i>Feature Work</i>	7
3.4	<i>Feature importance</i>	7
	<i>References</i>	9
	<i>Appendix</i>	10

1 Problem Statement

The growing accessibility of wearable tech has opened doors to continuously monitor various physiological factors. Detecting stress early has become pivotal, aiding individuals in proactively managing their health against the detrimental effects of prolonged stress exposure. Healthcare environments are inherently high stress, and nurses are particularly vulnerable due to demanding workloads and unpredictable patient needs. Early detection and understanding of stress patterns are critical for improving mental wellbeing, performance, and overall patient care.

This project aims to analyze and model physiological stress responses in nurses using wearable sensor data, including electrodermal activity (EDA), heart rate, skin temperature (TEMP), and motion signals. Each data point is labeled with a stress level: 0(low/no stress), 1(moderate stress), and 2(high stress). These labeled observations support supervised learning for stress prediction and allow for detailed analysis of temporal stress patterns and individual differences. The goal:

- 1) Predictive modeling: develop and evaluate machine learning models that classify stress levels of nurses based on physiological signals from wearable sensors.
- 2) Stress pattern analysis: Explore temporal patterns, individual differences, and the relative importance of physiological signals in relation to stress levels.

2 Methodology

The dataset contains wearable sensor data collected from nurses, including acceleration signals (X, Y, Z), electrodermal activity (EDA), heart rate (HR), skin temperature (TEMP), nurse identifier(id), timestamp (Datetime), and a stress label ranging from 0 to 2(low to high). The datetime column was converted to a standard datetime format.

2.1 Data Preprocessing

2.1.1 Motion Features Engineering

A composite motion variable was created from the three acceleration axes. After creating the motion variable, the original X, Y, Z columns were removed.

$$\text{Motion} = \sqrt{X^2 + Y^2 + Z^2}$$

To capture meaningful temporal behavior rather than instantaneous fluctuations, motion was also aggregated into 30 second rolling windows (grouped by nurse ID). For each window, the mean and variability of motion(std) were created to represent activity intensity and movement dynamics over time.

2.1.2 EDA Signal Processing

First, rows with EDA=0 were removed due to the absence of physiological signal. EDA data contain discontinuities due to gaps in sensor recording or variable sampling windows across nurses. To address this, the EDA time series was first split into continuous segment by detecting gaps greater than 500 ms between consecutive samples. Each continuous segment was then

processed independently to ensure the validity of the physiological decomposition. For segments longer than two seconds, EDA signals were decomposed using Neurokit2 function, which applies standard psychophysiological signal-processing algorithms. This method separates the raw EDA signal into:

- Tonic component (EDA_Tonic)
- Phasic component (EDA_Phasic)
- Skin Conductance Responses (SCR_Peaks)

Segments too short for reliable decomposition, or those processing errors, were assigned NAN values for all processed features. After processing, all segments were recombined, sorted by timestamp, and aligned with the main dataset. Missing values generated during signal processing were handled using forward and backward filling.

2.1.3 Heart Rate (HR) Feature Engineering

Heart rate values were grouped by nurse ID and timestamp, then processed using rolling 30 second windows to capture short term physiological variability. For each window, I computed:

- HR_mean(average HR)
- HR_std(short term variability)
- HR_diff (first order difference between consecutive HR measurements)

Rolling computations were applied to each nurse independently to avoid cross participant leakage. Instances with 0 values resulting from sensor dropout were forward filled to maintain continuity.

2.1.4 Temporal Feature Construction

To capture daily patterns, several time based features were extracted from the timestamp:

- Hour of day (0-23)
- Day of week (0=Monday to 6=Sunday)
- Date (calendar date without time)

To handle cyclical time, sin and cosine transformations were applied to improve model performance on temporal prediction tasks.

$$hour_sin = \sin\left(2\pi \frac{hour}{24}\right)$$

$$hour_cos = \cos\left(2\pi \frac{hour}{24}\right)$$

$$day_sin = \sin\left(2\pi \frac{day_of_week}{7}\right)$$

$$day_cos = \cos\left(2\pi \frac{day_of_week}{7}\right)$$

2.1.5 Handling Irregular Sampling and Resampling

Wearable data contained irregular sampling patterns across nurses: some nurses recorded data continuously, whereas others had sparse or intermittent recordings. The dataset was resampled to 30 second intervals, grouped by nurse ID, to standardize temporal spacing across all participants. However, due to large gaps between recording days for some nurses, resampling introduced extensive missing values corresponding to time ranges where no measurements were taken.

To preserve only valid physiological data, rows consisting entirely of resampled NaN values were removed. The 30 second resampling level was selected because it provided the best balance between reducing noise, improving comparability across nurses, and retaining sufficient density of data points for machine learning.

Table 1: Sampling frequency per ID

Daily or every 2 days	6B, 6D, 7E, F5, EG, 8B, DF
Twice a month	5C, 7A, E4
Every 3-7 days	15,83,94, BG, CE

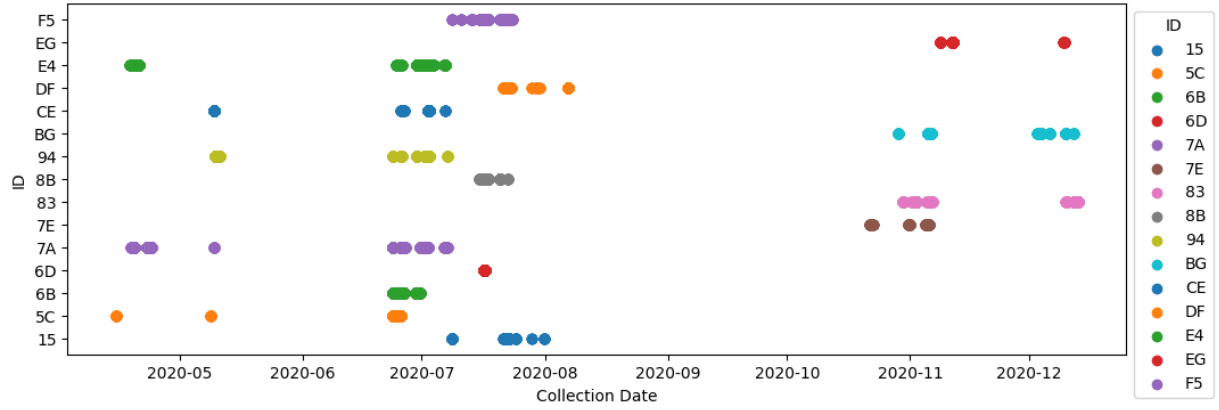


Figure 1: Sampling Pattern per Nurse

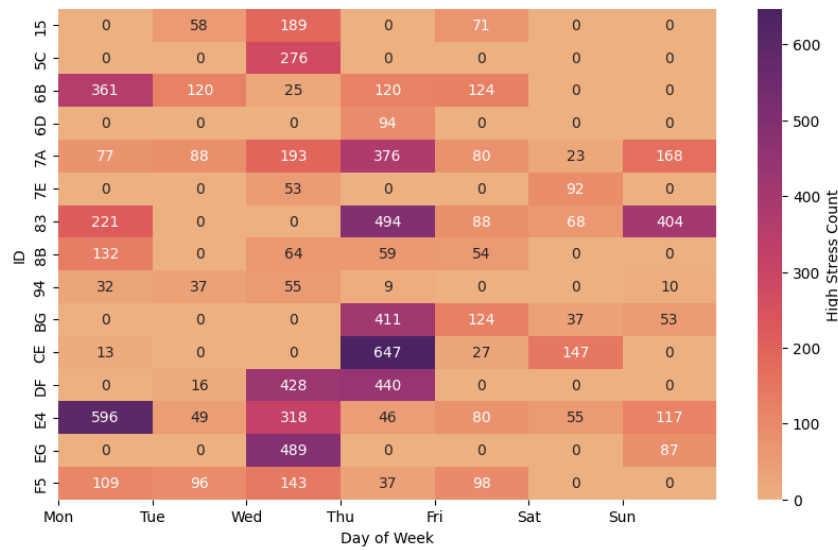


Figure2: High Stress (level2) Events Count per ID by Day of Week

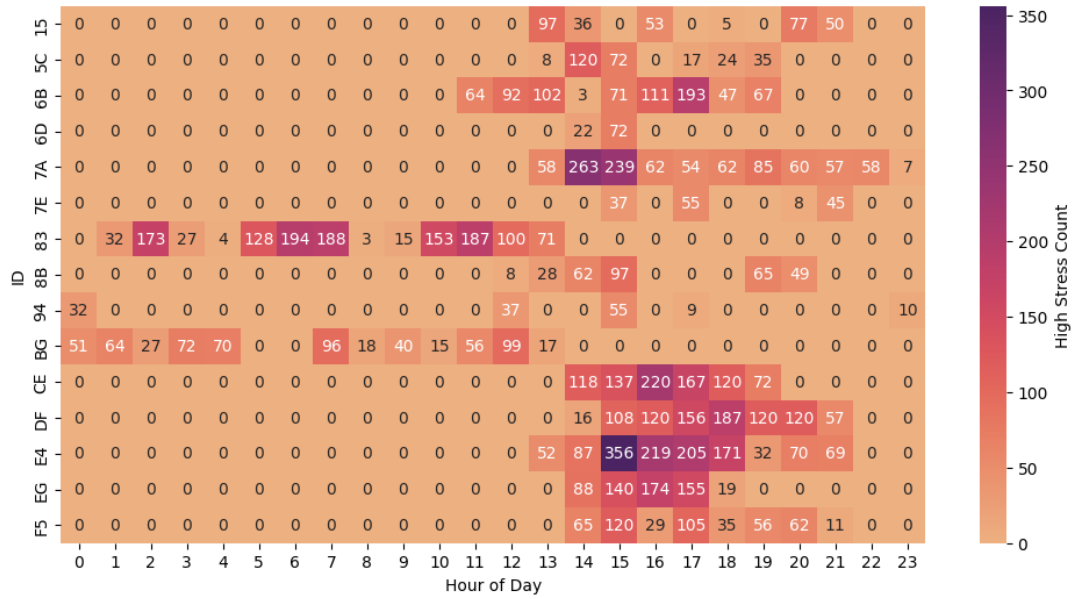


Figure 3: High Stress (level2) Events Count per ID by Hour of Day

To better understand when high-stress episodes occur, I visualized stress level across both the hour of day and day of week for each nurse. These plots highlight when individual nurses tend to experience elevated stress. Also, because presenting all 15 participants would overwhelm the main report, only the plots for ID 15 are included in the Appendix as a representative example: the remaining visualizations are available in the code section. [4,5,6,7]

2.2 Modeling

I removed id, datetime, date, hour, day of week columns that had no meaning to models or would cause data leakage to models. The dataset has only numerical variables. All these input features were standardized using StandardScaler before training. All models were trained using the same feature set. The dataset was split into 80% training and 20% testing sets and stratified by stress label. This dataset has substantial class imbalance. [8] Stress level 2 (high stress) was the most frequent label, while stress level 0 and especially stress level 1 were significantly underrepresented.

For handling class imbalance these strategies included:

- Class-weighted loss functions for models that support weighting
- SMOTE oversampling for minority classes in the training set.

Models

Key hyperparameters I set; n_estimators, max_depth, random_state, class_weight
All models were evaluated with 5-fold time series cross-validation.

- Random Forest; 300 trees, unlimited depth
- XGBoost; multiclass objective, eval_metric='mlogloss'
- LightGBM; multiclass objective
- Gradient Boosting Classifier; learning rate:0.5, max_depth:5, early_stopping
- Logistic Regression; multinomial, max_iter=1000

3 Results and Evaluation

3.1 Results

Model	SMOTE	Accuracy	Macro F1	Class 0-F1	Class 1-F1	Class 2- F1
Random Forest	No	0.98	0.96	0.95	0.93	0.99
Random Forest	Yes	0.98	0.97	0.96	0.96	0.99
XGBoost	No	0.69	0.34	0.20	0.00	0.81
XGBoost	Yes	0.98	0.97	0.97	0.96	0.99
LightGBM	No	0.71	0.34	0.21	0.00	0.82
LightGBM	Yes	0.98	0.97	0.96	0.97	0.99
Gradient Boosting	No	0.72	0.33	0.16	0.00	0.83
Gradient Boosting	Yes	0.89	0.83	0.81	0.75	0.93
Logistic Regression	No	0.55	0.46	0.46	0.28	0.66
Logistic Regression	Yes	0.55	0.47	0.47	0.29	0.65

3.2 Evaluation

I evaluated the performance of 5 models on the test set, both with and without SMOTE oversampling. Random Forest, XGBoosting, and LightGBM models achieved the highest overall performance, with accuracy and macro-f1 scores above 0.97 after SMOTE.

Logistic regression performed poorly, even with oversampling, confirming that the relationships between physiological and temporal features and stress levels are highly non linear.

Oversampling using SMOTE substantially improved minority class performance. Particularly for stress level 1 which is underrepresented in the dataset.

Random Forest performed strongly even without oversampling, this shows that tree based ensemble methods naturally handle noise and feature interactions in physiological data.

However, after applying SMOTE, XGBoost and LightGBM performed as good as Random Forest, indicating that gradient boosting methods benefit more from balanced training data.

3.3 Feature Work

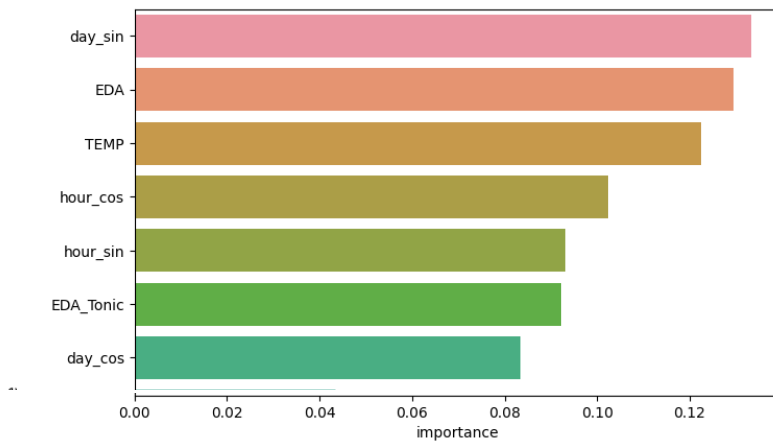
All models were trained and evaluated using mixed subject data, meaning that data from all nurses contributed to both the training and testing sets. A full cross-subject evaluation (leave-one-out) was not performed in this study. This would be a valuable next step to assess the generalizability of stress detection to unseen individuals.

Probability calibration was not included in the analysis because the primary focus was on classification accuracy and minority-class performance. This would be an important next step if the system is intended to generate risk alerts or provide continuous stress monitoring.

3.4 Feature importance

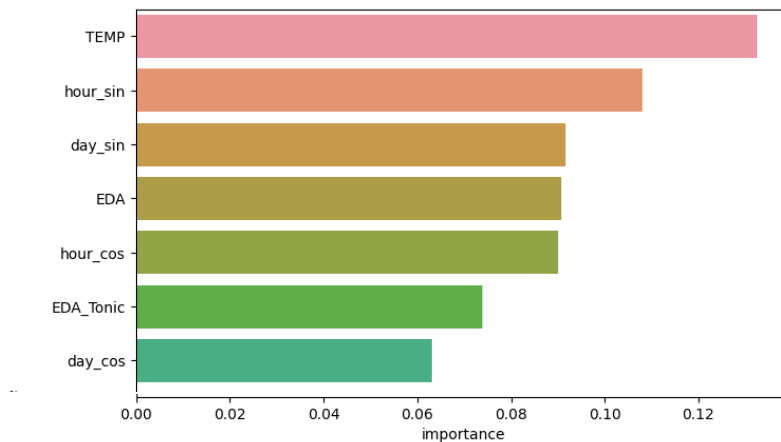
To understand which variables, contribute most to stress detection, feature importance was examined for the best performing models: Random Forest, XGBoost and LightGBM.

Random Forest



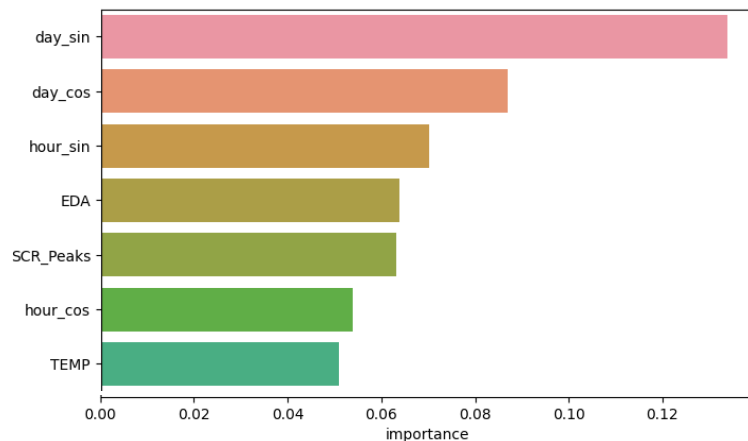
This pattern indicates that the Random Forest placed strong weight on circadian and weekly rhythms, alongside EDA (sweat) based stress responses and temperature, suggesting a blend of physiological and temporal influences on stress.

LightGBM



This pattern indicates that the LightGBM placed even more strong weight on time of day and day of week encodings, indicating that temporal structure plays a key role in how stress manifests throughout a shift or workday. EDA (sweat) remained one of the most influential physiological features.

XGBoosting



Compared to Random Forest and LightGBM, XGBoost assigned higher importance to SCR_Peaks, which aligns with the model's ability to detect subtle non linear patterns related to sympathetic arousal. Temporal encodings again appeared prominently. Reinforcing that daily/weekly cycles strongly influence nurse stress patterns.

References

- [1] Makowski, D., Pham, T., Lau, Z.J. *et al.* NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behav Res* **53**, 1689–1696 (2021). <https://doi.org/10.3758/s13428-020-01516-y>
- [2] Han, H., Wang, WY., Mao, BH. (2005). Borderline-SMOTE: A New Over-Sampling Method in Imbalanced Data Sets Learning. In: Huang, DS., Zhang, XP., Huang, GB. (eds) *Advances in Intelligent Computing. ICIC 2005. Lecture Notes in Computer Science*, vol 3644. Springer, Berlin, Heidelberg. https://doi.org/10.1007/11538059_91
- [3] Chen, Tianqi and Carlos Guestrin. "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2016): n. pag.
- [4] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Neural Information Processing Systems*.
- [5] Jukiewicz, M., & Marcinkowska, J. (2025). Analysis of Electrodermal Signal Features as Indicators of Cognitive and Emotional Reactions-Comparison of the Effectiveness of Selected Statistical Measures. *Sensors (Basel, Switzerland)*, 25(11), 3300.

Appendix

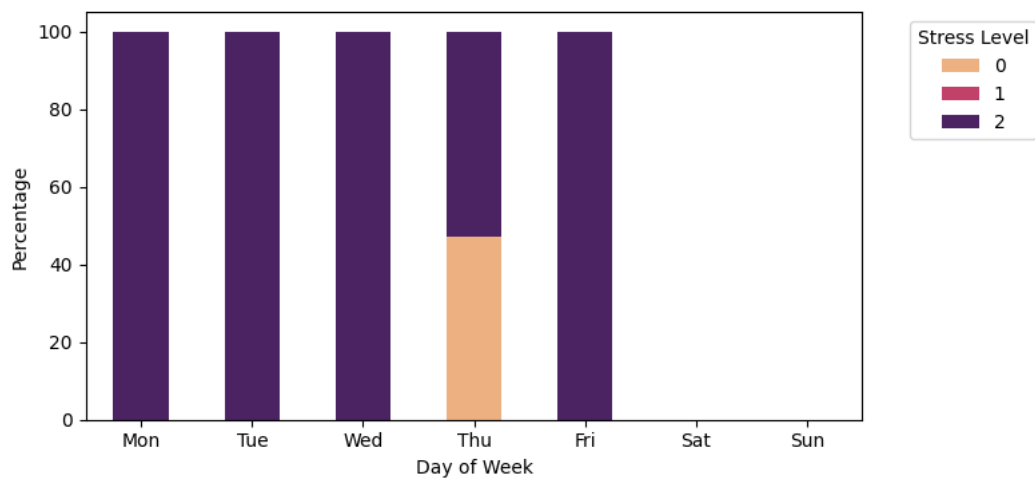


Figure 4: Percentage of Stress Levels per Day – ID 15

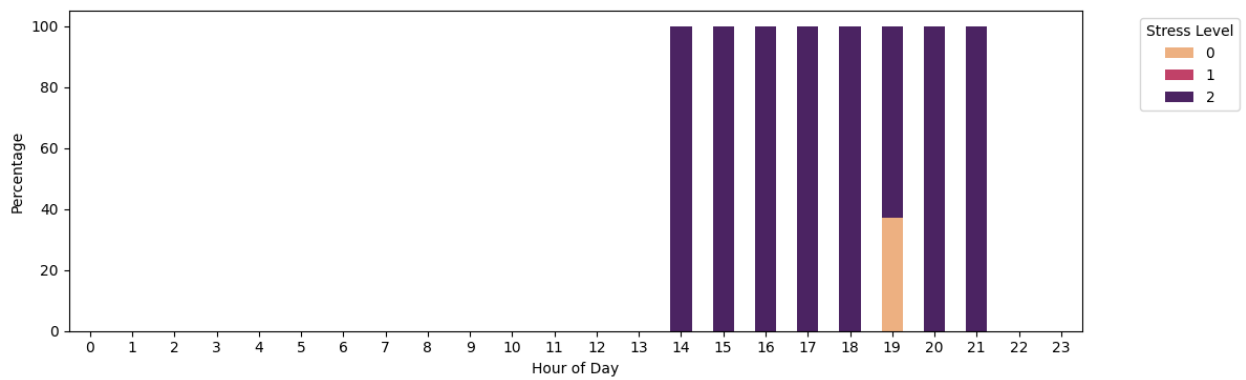


Figure 5: Percentage of Stress Level per Hour- ID 15

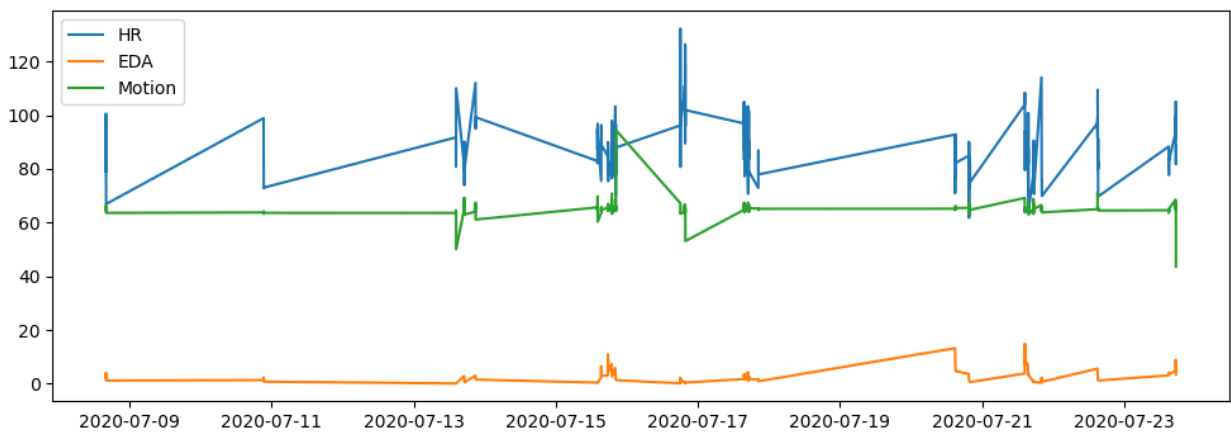


Figure 6: Physiological signals for ID 15

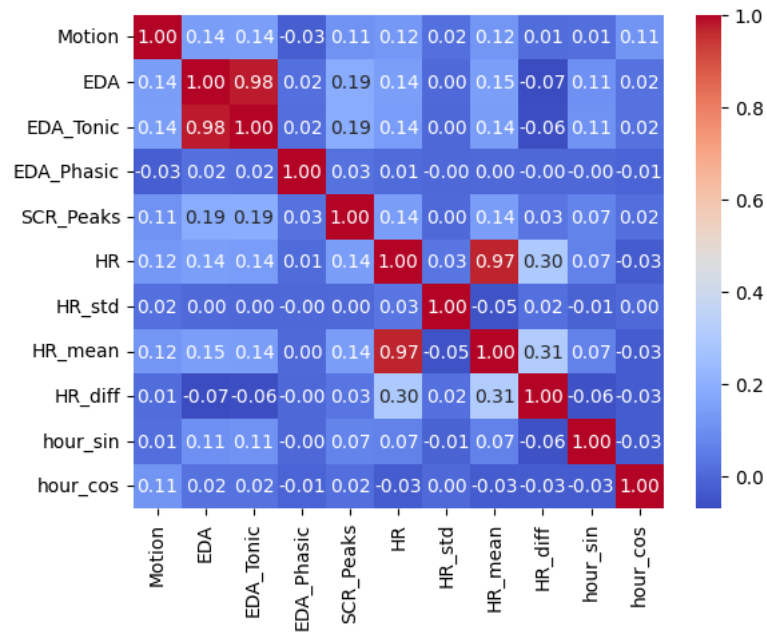


Figure 7: Correlation plot for Physiological signals

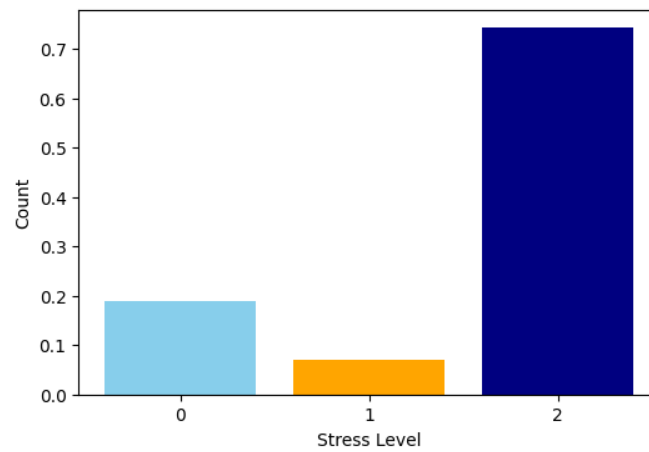


Figure 8: Distribution of Stress Levels in the Nurse dataset